Principles of Data Visualization

BI304

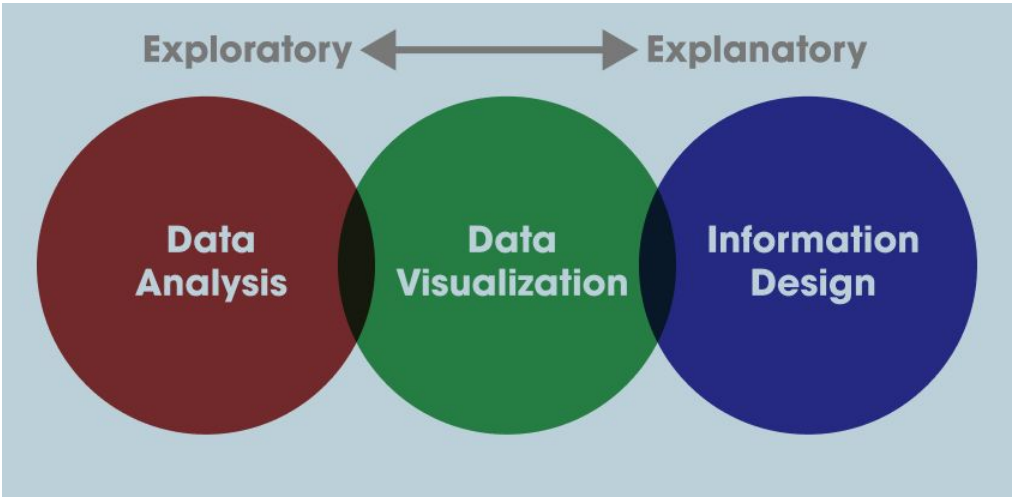
Noptanit Chotisarn
Thammasat Business School

Data Visualization

- Data visualization has been defined as “the use of visual representations to explore, make sense of, and communicate data”. And this usually means is information visualization.
- Because information is the aggregation, summarization, and contextualization of data (raw facts), what is portrayed in visualizations is the information, not the data.

Data visualization (cont.)

- Data visualization is closely related to the fields of information graphics, information visualization, scientific visualization, and statistical graphics. Nowadays, the major forms of data visualization available in applications that used to create scorecards and dashboards.



A Brief History of Data Visualization

(and the role of libraries and librarians)



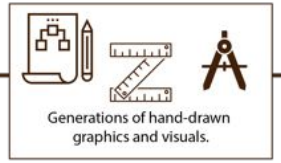
1914

Graphic Methods for Presenting Facts

by Willard C. Brinton

The principles for a grammar of graphic presentation are so simple that a remarkably small number of rules would be sufficient to give a universal language.

Willard C. Brinton



Generations of hand-drawn graphics and visuals.

1977



The personal computer arrives.

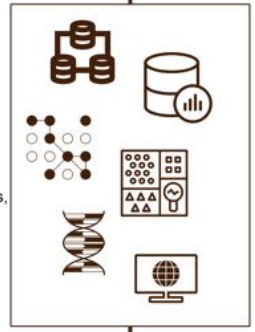
2017

A special issue of the **Journal of eScience Librarianship** is published, highlighting the future of librarians, librarianship, and data visualization.



2000 'til today

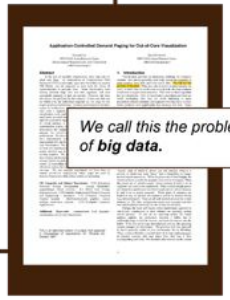
Rapid expansion of data generating, data analysis, data sequencing, and data visualizing tools. Access, management, and preservation of data becomes a key concern and disciplines emerge to address these different aspects related to data including **libraries and librarians.**



1997

Application-Controlled Demand Paging for Out-of-Core Visualization

by Michael Cox & David Ellsworth



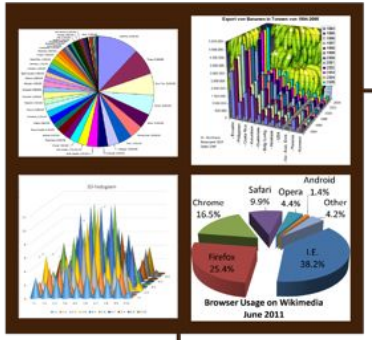
*We call this the problem of **big data.***

1987



Microsoft Excel: Spreadsheet software for everyone.

1990 and beyond

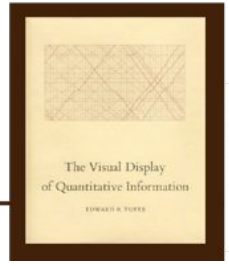


The explosion of "CHARTJUNK."

Having the means to create graphs with a computer doesn't guarantee that we'll do it effectively any more than having word processing software makes us great writers.

Stephen Few

1983, 1990, 1997...



It is not how much empty space there is, but rather how it is used. It is not how much information there is, but rather how effectively it is arranged.

Edward Tufte

Gore, S. A. (2018). A Brief History of Data Visualization (and the role of libraries and librarians). Journal of eScience Librarianship, 7(1), 6.

Different Types of Charts and Graphs

- Often end users of business analytics systems are not sure what type of chart or graph to use for a specific purpose.
- Some charts or graphs are better at answering certain types of questions. Some look better than others.
- Some are simple; some are rather complex and crowded. What follows is a short description of the types of charts and/or graphs commonly found in most business analytics tools and the types of questions they are better at answering/analyzing.

This material is compiled from several published articles and other literature (Abela, 2008; Hardin et al., 2012; SAS, 2014).

Basic Charts and Graphs

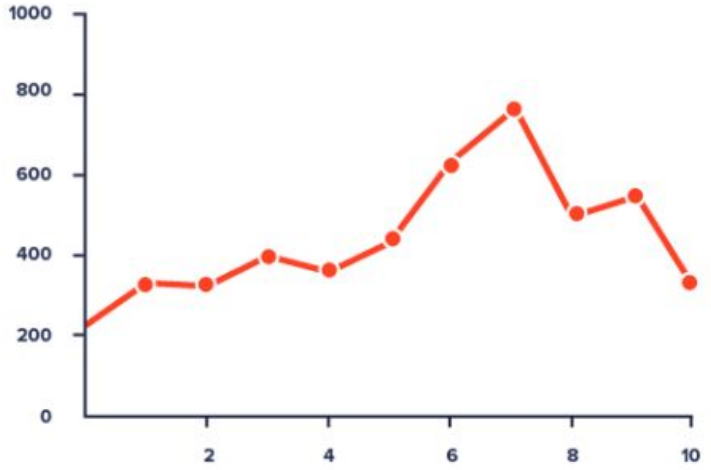
- What follows are the basic charts and graphs that are commonly used for information visualization.
  - LINE CHART
  - BAR CHART
  - PIE CHART
  - SCATTER PLOT
  - BUBBLE CHART

Specialized Charts and Graphs

- The graphs and charts that we review in this section are either derived from the basic charts as special cases or they are relatively new and are specific to a problem type and/ or an application area.
  - HISTOGRAM
  - GANTT CHART
  - PERT CHART
  - GEOGRAPHIC MAP
  - BULLET GRAPH
  - HEAT MAP
  - HIGHLIGHT TABLE
  - TREE MAP

Line Chart (Line Graph)

- The line chart is perhaps the most frequently used graphical visuals for time-series data.
- Line charts show the relationship between two variables; they are most often used to track changes or trends over time (having one of the variables set to time on the x-axis).



Family
- Plot

Function
- Distribution
- Trend over time

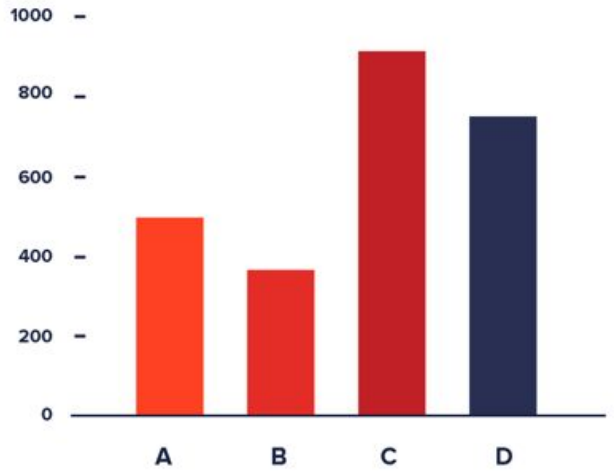
Input

X	Y
1	30
2	34
3	38

Ref : https://datavizproject.com

Bar Chart

- The bar chart is among the most basic visuals used for data representation.
- They are effective when you have nominal data or numerical data that split nicely into different categories so you can quickly see comparative results and trends within your data.
- These are often used to compare data across multiple categories.



Family

- Chart

Function

- Comparison
- Trend over time

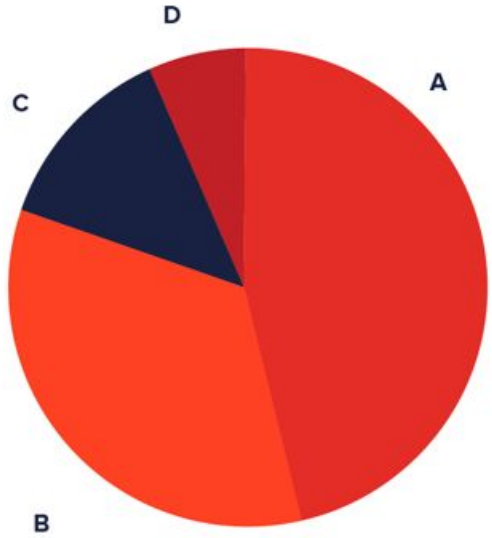
Input

A	14
B	16
C	12

Ref : https://datavizproject.com

Pie Chart

- The pie chart are often incorrectly used.
- Pie charts should be used only to illustrate relative proportions of a specific measure.
- If the number of categories to show is more than just a few (say more than four), one should seriously consider using a bar chart instead of a pie chart.



Family
- Chart

Function
- Comparison

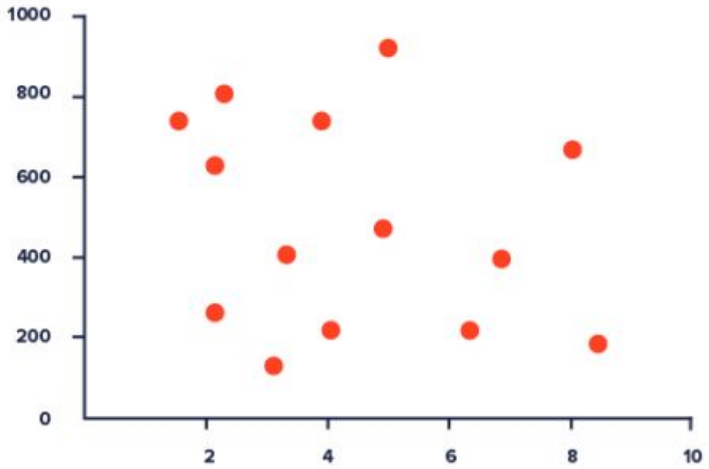
Input

A	32%
B	40%
C	28%

Ref : https://datavizproject.com

Scatter Plot

- The scatter plot is often used to explore the relationship between two or three variables (in 2D or 3D visuals).
- Scatter plots are an effective way to explore the existence of trends, concentrations, and outliers.
- The data is displayed as a collection of points, each having the value of one variable determining the position on the horizontal axis and the value of the other variable determining the position on the vertical axis.



Family
- Plot

Function
- Correlation
- Distribution

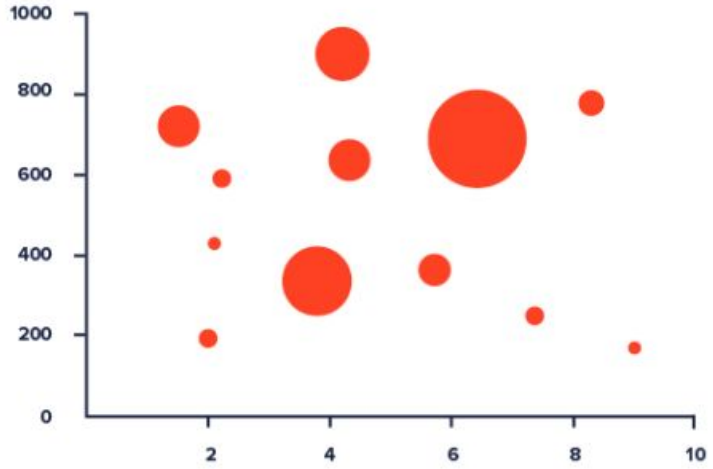
Input

	X	Y
A	30	14
B	34	16
C	38	10

Ref : https://datavizproject.com

Bubble Chart

- The bubble chart is often an enhanced version of scatter plots.
- Bubble charts are not a new visualization type they should be viewed as a technique to enrich data illustrated in scatter plots (or even geographic maps). By varying the size and/or color of the circles, one can add additional data dimensions, offering more enriched meaning about the data.



Family
- Chart
- Plot

Function
- Comparison
- Distribution
- Trend over time

Input

X	Y	Z
1	30	10
2	34	14
3	38	12
	⌄	

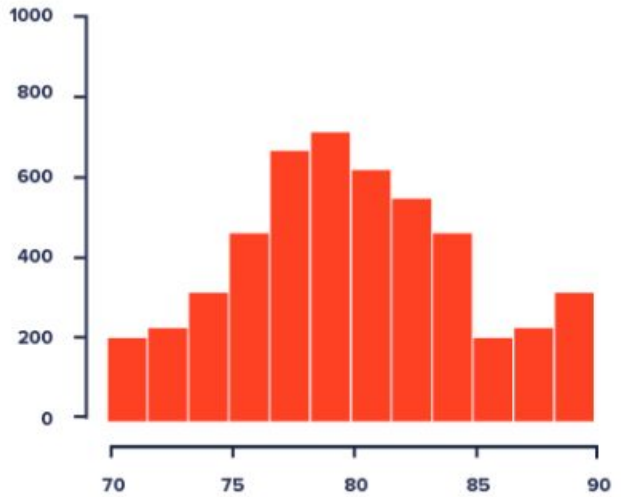
Ref : https://datavizproject.com

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  - PERT CHART
  - GEOGRAPHIC MAP
  - BULLET GRAPH
  - HEAT MAP
  - HIGHLIGHT TABLE
  - TREE MAP

Histogram

- Graphically speaking, a histogram looks just like a bar chart. The difference between histograms and generic bar charts is the information that is portrayed.
- Histograms are used to show the frequency distribution of one variable or several variables. In a histogram, the x-axis is often used to show the categories or ranges, and the y-axis is used to show the measures/values/frequencies.
- Histograms show the distributional shape of the data. That way, one can visually examine whether the data are normally or exponentially distributed.



Family
- Chart

Function
- Distribution

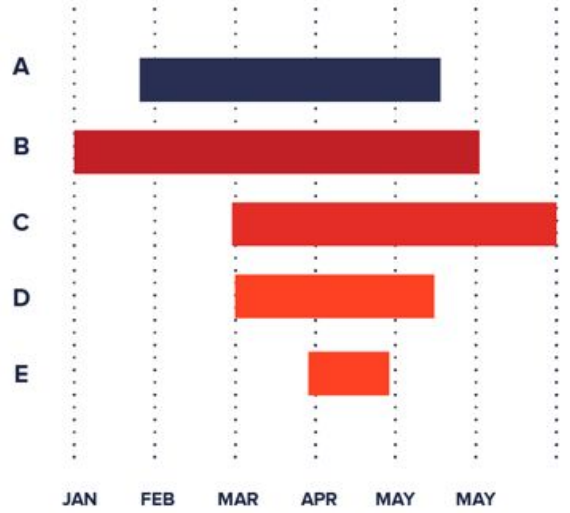
Input

X	Y
0-2	30
2-4	34
4-6	38

Ref : https://datavizproject.com

Gantt Chart

- A Gantt chart is a special case of horizontal bar charts used to portray project timelines, project tasks/activity durations, and overlap among the tasks/activities. By showing start and end dates/times of tasks/activities and the overlapping relationships, Gantt charts provide an invaluable aid for management and control of projects.



Family

- Chart

Function

- Distribution

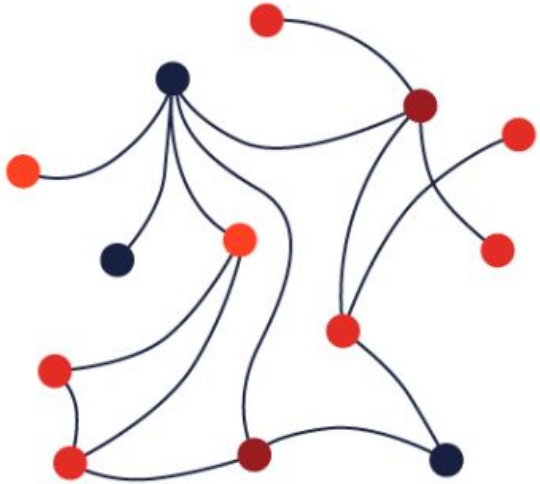
Input

	Start time	End time
A	1-4-2015	6-4-2015
B	10-4-2015	18-4-2015
C	12-4-2015	20-4-2015

Ref : https://datavizproject.com

PERT Chart

- This chart (also called a network diagram) is developed primarily to simplify the planning and scheduling of large and complex projects.
- A PERT chart shows precedence relationships among project activities/tasks.It is composed of nodes (represented as circles or rectangles) and edges (represented with directed arrows).
- Based on the selected PERT chart convention, either nodes or the edges can be used to represent the project activities/tasks (activity-on-node versus activity-on-arrow representation schema).



Family
- Diagram

Function
- Concept visualisation
- Correlation

Input

	A	B	C
A		1	1
B	1		0
C	0	0	

Ref : https://datavizproject.com

Bullet Graph

- A bullet graph is often used to show progress toward a goal. This graph is essentially a variation of a bar chart.
- Bullet graphs were developed to overcome the fundamental issues of gauges and meters: they typically display too little information, require too much space, and are cluttered with useless and distracting decoration



Family
- Chart
- Plot

Function
- Part to whole

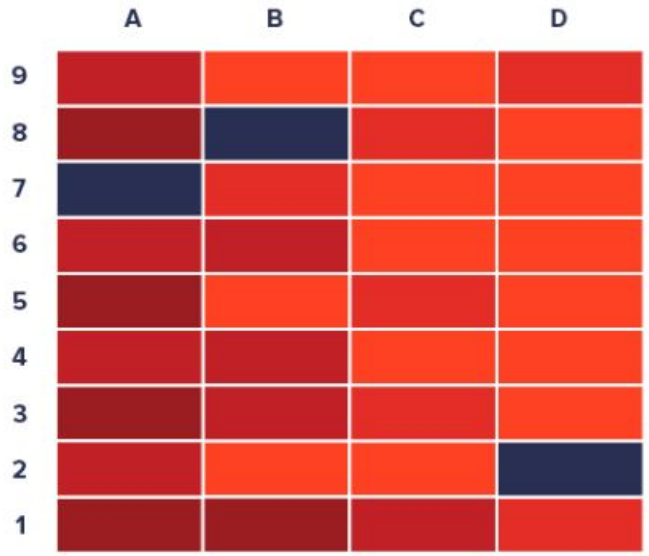
Input

A
60%

Ref : https://datavizproject.com

HEAT MAP

- The heat map, or Heat Table, Density Table, heat map, is a great visual to illustrate the comparison of continuous values across two categories using color.
- The goal is to help the user quickly see where the intersection of the categories is strongest and weakest in terms of numerical values of the measure being analyzed.



Family
- Chart
- Table

Function
- Comparison
- Correlation
- Distribution
- Trend over time

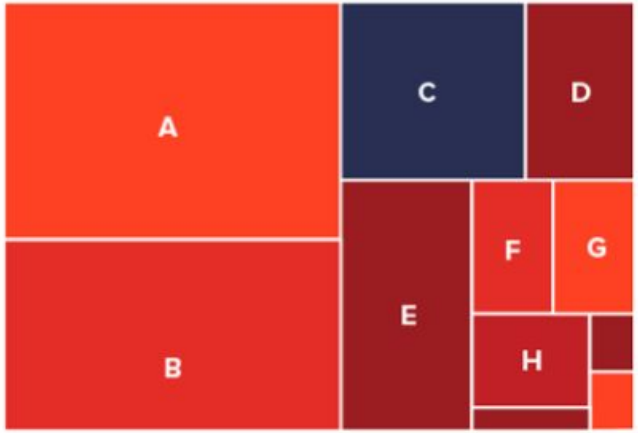
Input

1	20%	50%
2	10%	40%
3	30%	70%

Ref : https://datavizproject.com

TREE MAP

- Treemaps display hierarchical (tree-structured) data as a set of nested rectangles. Each branch of the tree is given a rectangle, which is then tiled with smaller rectangles representing sub-branches.
- A leaf node’s rectangle has an area proportional to a specified dimension on the data. Often the leaf nodes are colored to show a separate dimension of the data.



Family
- Chart

Function
- Comparison
- Distribution

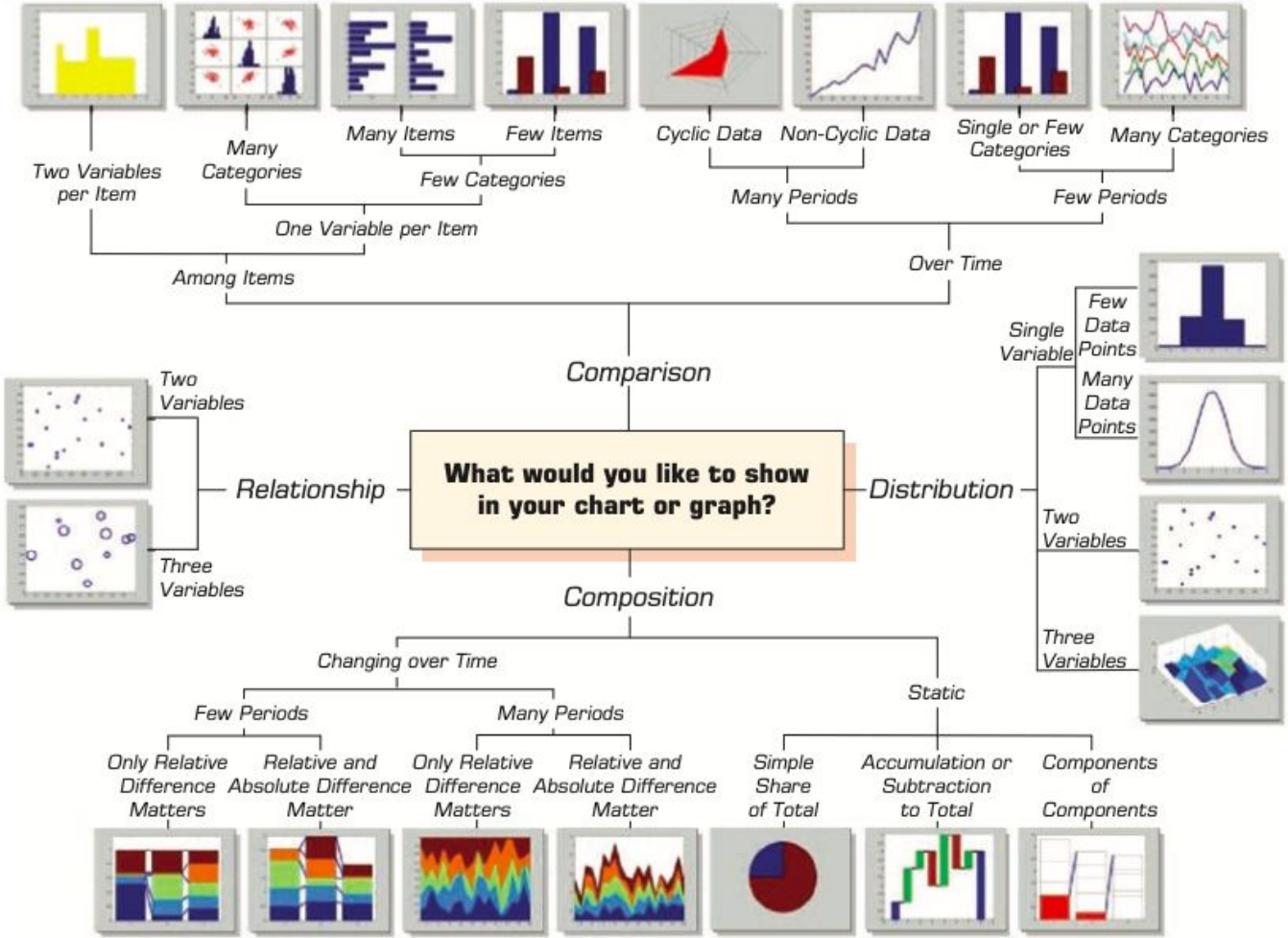
Input

A	A_1	14
	A_2	16
B	B_1	10
	⌄	

Ref : https://datavizproject.com

Which Chart or Graph Should You Use?

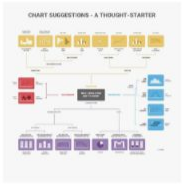
- There is not one best chart or graph because if there were, we would not have so many chart and graph types. They all have somewhat different data representation “skills.” Therefore, the right question should be, ***“Which chart or graph is the best for a given task?”***
- The capabilities of the charts given in the previous section can help in selecting and using the proper chart/graph for a specific task, but doing so still is not easy to sort out. Several different chart/graph types can be used for the same visualization task. One rule of thumb is to select and use the simplest one from the alternatives to make it easy for the intended audience to understand and digest.



A Taxonomy of Charts and Graphs.

Source: Adapted from Abela, A. (2008). Advanced Presentations by Design: Creating Communication That Drives Action. New York: Wiley.

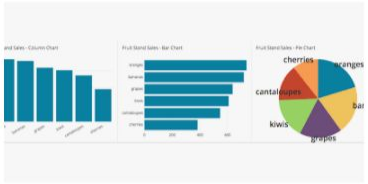
decision tree | visualization design | kpi | data analysis descriptive diagnostic predictive | chart | graph | diagram | visualization types | visualization tools | scatter plot


Data Visualization - tell a sto...
big2smart.com


Multiple views on how ...
medium.com



5 Quick and Easy Data Visualizations in ...
kdnuggets.com



How to Select a Data Visualization | Tutorial by Chartio
chartio.com

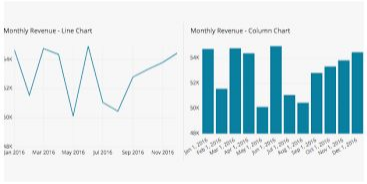

How to Choose the Right Visualization for Your ...
logianalytics.com



Data Visualization 101: How to Choose a Chart Typ...
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Choosing The Right Data Visualization ...
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How to Select a Data Visualization | Tutorial by Chartio
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predictiveanalyticstoday.com



How to Choose the Right Data Visualization | Tutorial by Chart...
chartio.com


How to Select a Data Visualizati...
blogs.starcio.com

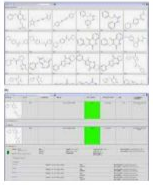


Trump Learning
trumplearning.com


Creating visualizations and graphs
docs.looker.com



5 Steps To Choosing Great Data Visu...
towardsdatascience.com


Data visualization. Hit ...
researchgate.net

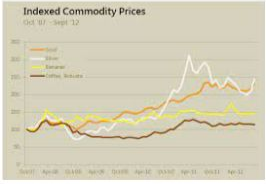


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treehousetechgroup.com


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The Best Data Visualization Tools (Select Right Techniques & ...
data-nectar.com



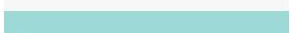
Selecting the Right Data Visualizations | Pikt...
piktochart.com


Datavisualization.ch Selected Tools
selection.datavisualization.ch










Recommended

https://www.data-to-viz.com/

Data visualization is about understanding

1. Think about your target audience
2. Don't assume knowledge
3. Serve the setting as well as the audience
4. Define what success looks like
5. Remember, You can't please everyone
6. Decide what you want to show
7. Then determine how to show it
8. The annotation layer is the most important
9. Layer information to suit the context
10. Use colour wisely

Context

- In order for creators to facilitate understanding they should be designing the content for the consumers - for someone else.
- The most important thing as a creator is to avoid being your own audience. You know too much about the subject already and you'll just end up summing up your own understanding.
- Using personas is a good way to avoid this. In essence try to think of a typical person who might consume the work - then design with them in mind throughout. Even better, speak to some real users and find out what they want.

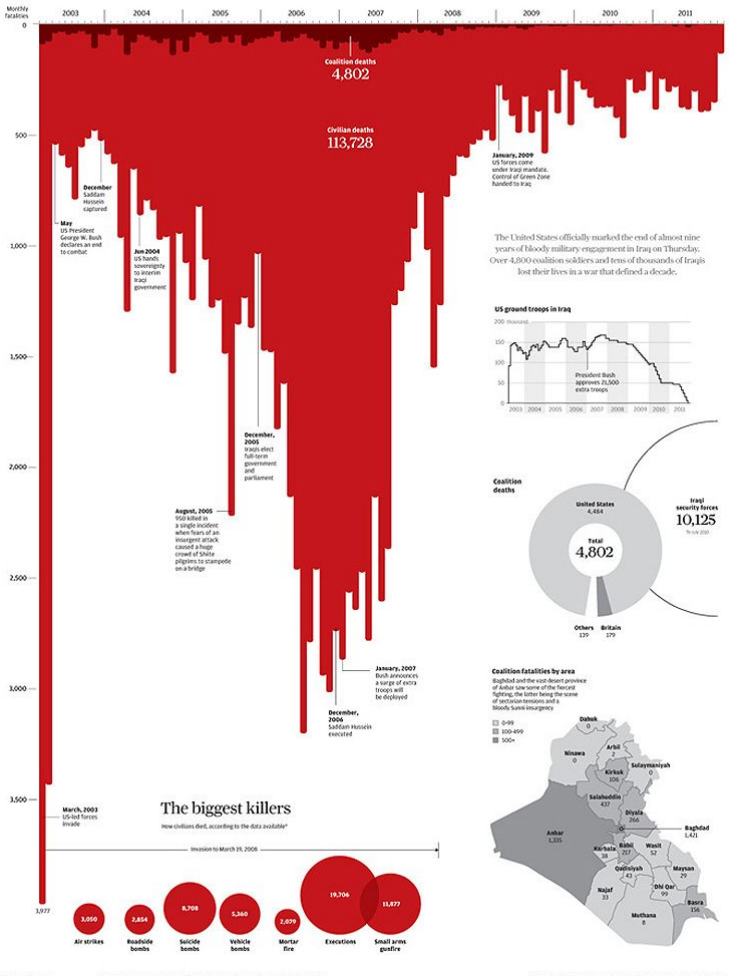
Purpose

The aims and objectives of your data visualization. What are you trying to achieve?

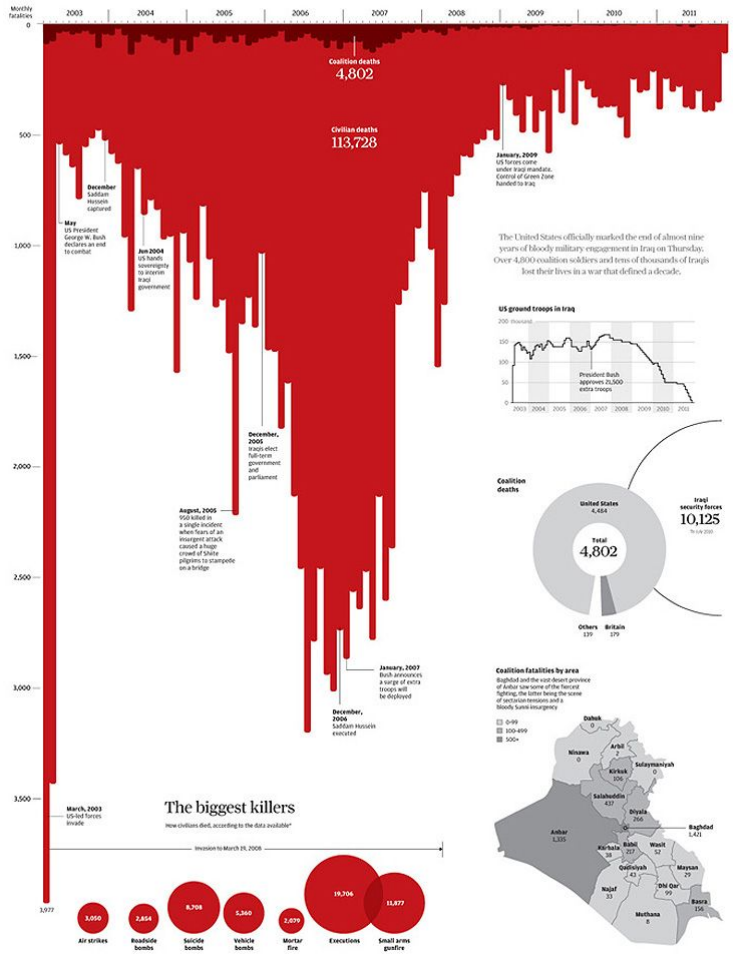
- Explaining to educate the reader?
- Informing to help decision-making?
- Shocking to change behavior?

As a producer, you need to think carefully about what it is that you want the reader to take away from your data visualization.

Case Study


Monthly fatalities
2003
2004
2005
2006
2007
2008
2009
2010
2011
Coalition deaths
4,802
Civilian deaths
113,728
January, 2009
US forces come under Iraqi mandate. Control of Green Zone handed to Iraq
May
US President George W. Bush declares an end to combat
December
Saddam Hussein captured
Jun 2004
US hands sovereignty to interim Iraqi government
August, 2005
950 killed in a single incident when fears of an insurgent attack caused a huge crowd of Shiite pilgrims to stampede on a bridge
December, 2005
Iraqis elect full-term government and parliament
January, 2007
Bush announces a surge of extra troops will be deployed
December, 2006
Saddam Hussein executed
March, 2003
US-led forces invade
3,977
The United States officially marked the end of almost nine years of bloody military engagement in Iraq on Thursday. Over 4,800 coalition soldiers and tens of thousands of Iraqis lost their lives in a war that defined a decade.
US ground troops in Iraq
200 thousand
150
100
50
0
President Bush approves 21,500 extra troops
Coalition deaths
United States
4,484
Iraqi security forces
10,125
Total
4,802
Others
139
Britain
179
Coalition fatalities by area
Baghdad and the vast desert province of Anbar saw some of the fiercest fighting, the latter being the scene of sectarian tensions and a bloody Sunni insurgency
0-99
100-499
500+
Dahuk
0
Arbil
2
Ninawa
0
Kirkuk
106
Solaymaniyah
0
Salahaddin
437
Diyala
266
Baghdad
1,421
Anbar
1,335
Karbala
38
Babil
97
Wasit
52
Qadisiyah
43
Maysan
29
Najaf
33
Dhi Qar
99
Basra
156
Muthana
8
The biggest killers
How civilians died, according to the data available*
Invasion to March 19, 2008
3,050
Air strikes
2,854
Roadside bombs
8,708
Suicide bombs
5,360
Vehicle bombs
2,039
Mortar fire
19,706
Executions
11,877
Small arms gunfire

Iraq's bloody toll

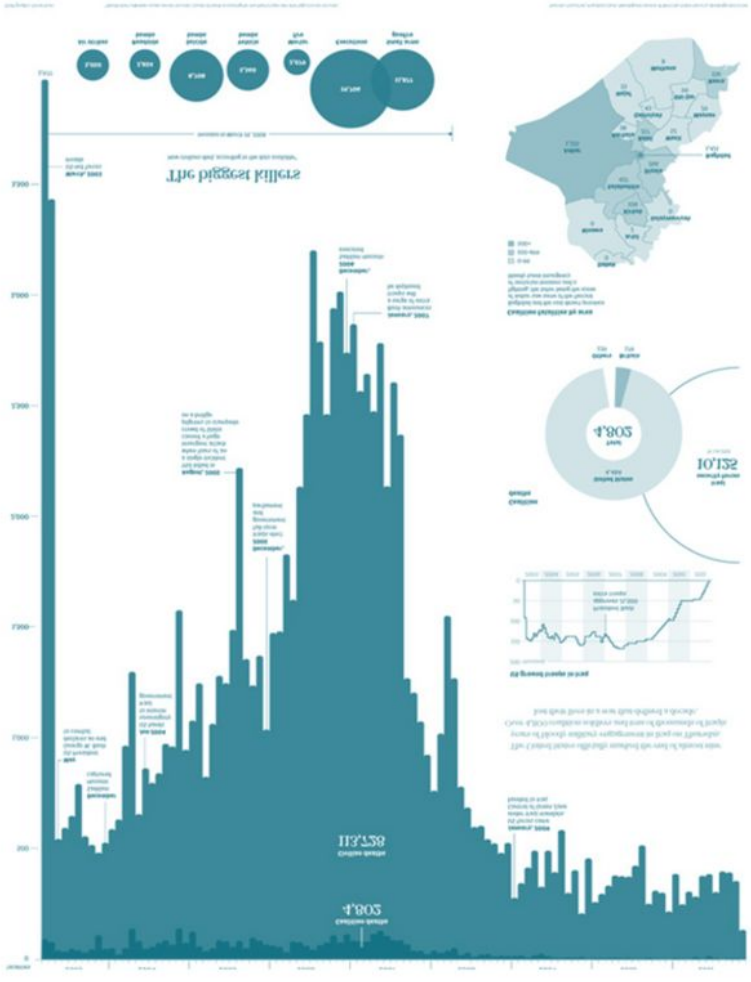


SCMP graphic: Simon Scarr

*Deaths from unknown causes are not included. Causes of death accounting for less than 0.5 per cent of all registrations are not used.

Sources: Consolidated Iraq Body Count, New England Journal of Medicine, Global Security, Brookings Institute

Iraq: Deaths on the decline



Design

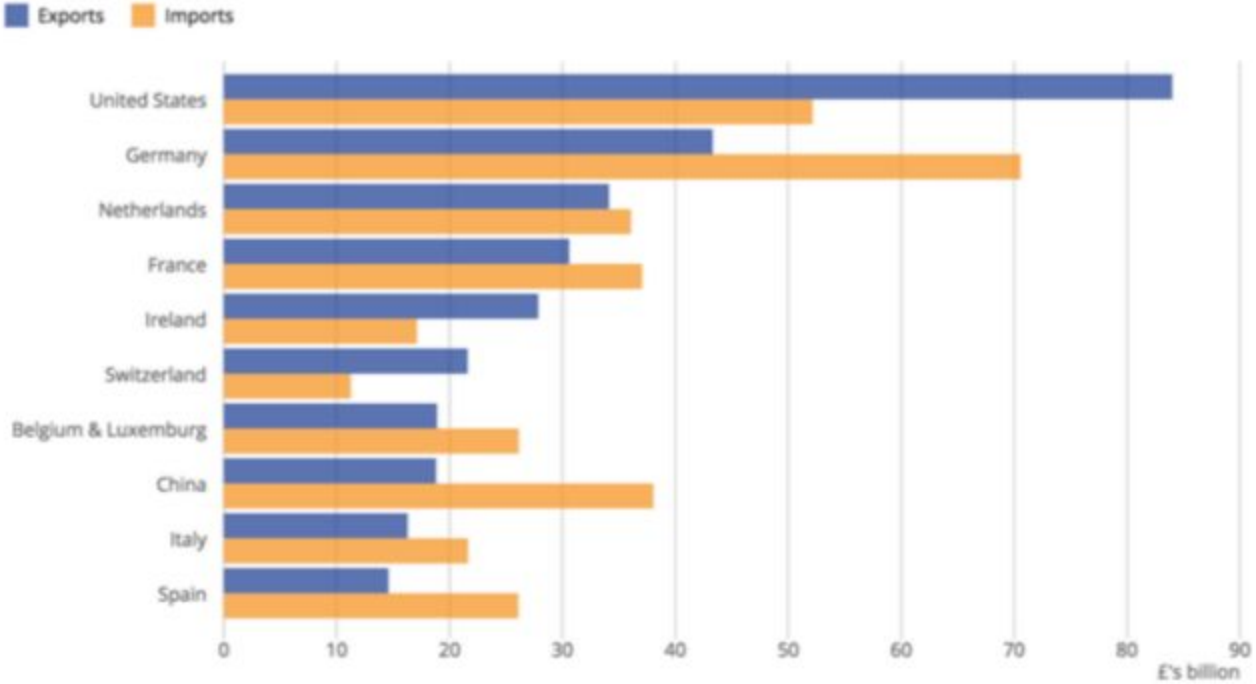
- Most of the judgements you will have to make will be in the design of the charts.
- By now we should know our Who, Where and When (the audience and setting i.e. the context). We are clear on the Why (the purpose of the visualization) and we have acknowledged the constraints imposed on us, so armed with this knowledge we can consider our design (i.e. what to show and how to show it).
- We have through personas, thinking about the need to define and serve your target audience. This is user-centered design.
- User-centered design is about empathy. It's not enough to rely on your own instincts and be driven by your own tastes. The user need is what is important.
- How do we best facilitate understanding and make the information useful?
- Data visualization design is a massive topic and we are not going to cover everything, but we will use case studies and examples to illustrate some important points.

http://ft-interactive.github.io/visual-vocabulary/

Case Study

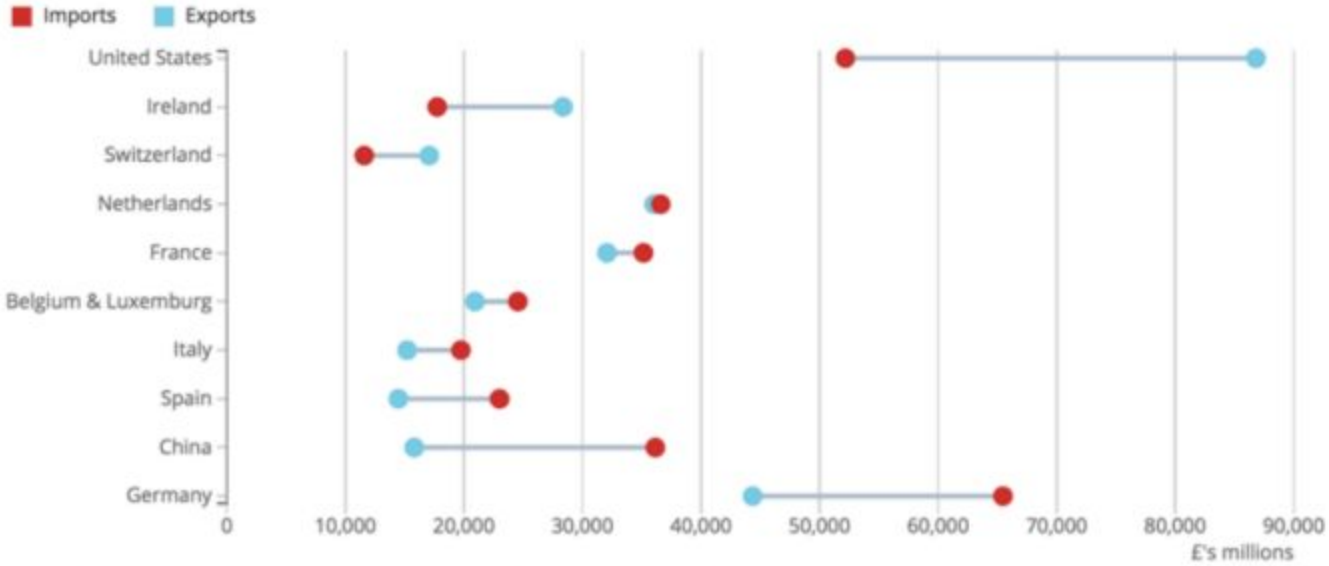
- This visualization relate to a project before the EU referendum to give a picture of UK trade with the world and Europe.
- One section of this analysis was looking at trade relationships with other countries, in particular our top 10 trading partners.
- The important thing was WHAT they wanted to show - they wanted to give the ability to see the magnitude of imports and exports but also the difference between imports and exports.
- In this example the angle was the size and balance of trade in and out of the UK, the framing was the top 10.

Top 10 trading partners in goods and services, current prices, 2014



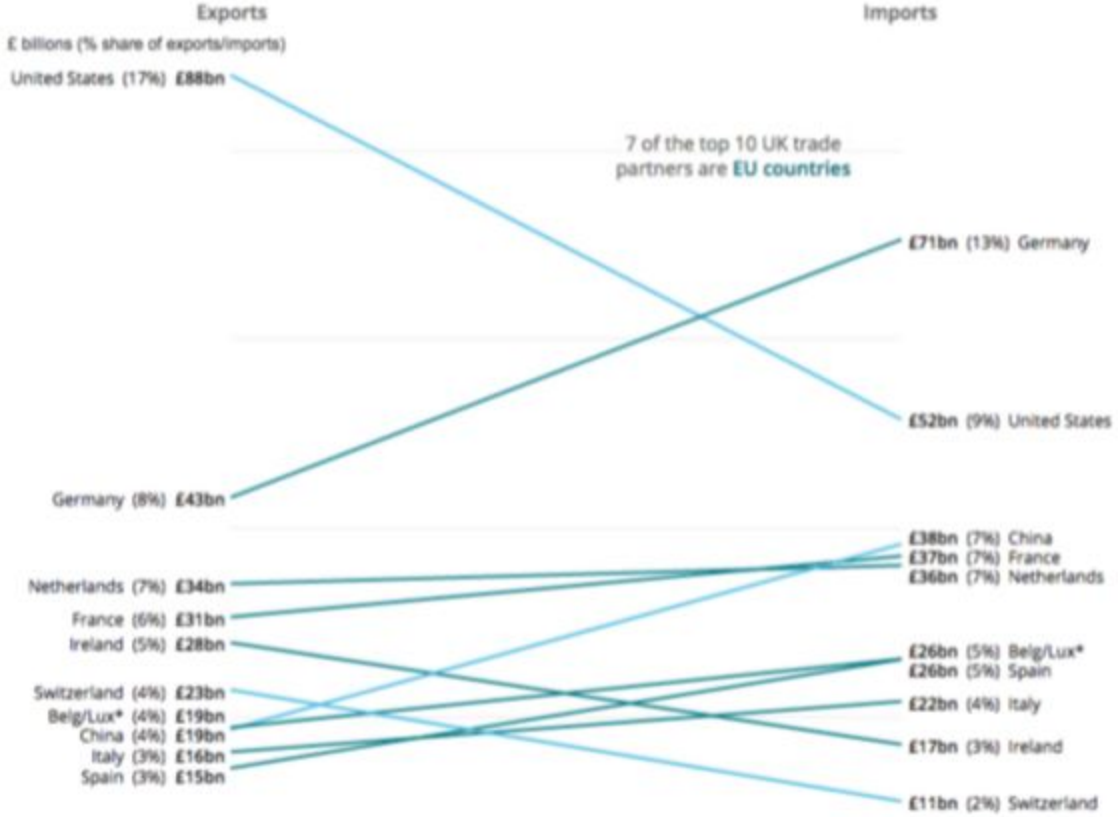
Source: Pink book (Geographical break down of the current account)

Top 10 trading partners in goods and services, current prices, 2014



Source: Pink Book (Geographic break down of the current account)

Top 10 trading partners in goods and services, current prices, 2014



Case Study - Table

https://dataviztraining.dwpdata.info/img/Tables.gif

Case Study - Graphs

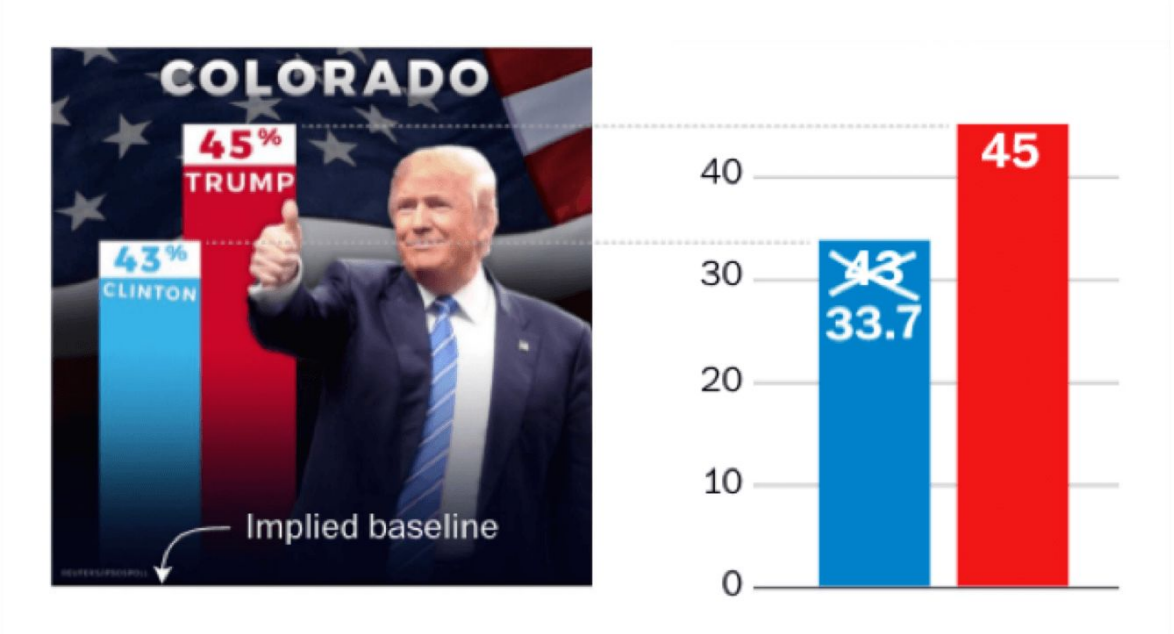
https://dataviztraining.dwpdata.info/img/DWP_CS1.gif

https://dataviztraining.dwpdata.info/img/DWP_CS2.gif

Case Study - Bar Charts



Case Study - Bar Charts



Case Study - Bar Charts

https://dataviztraining.dwpdata.info/img/DWP_CS3.gif

Case Study - Line Charts

https://dataviztraining.dwpdata.info/img/linechart.gif

Case Study - Line Charts (Small and Multiples)

https://dataviztraining.dwpdata.info/img/DWP_CS5.gif

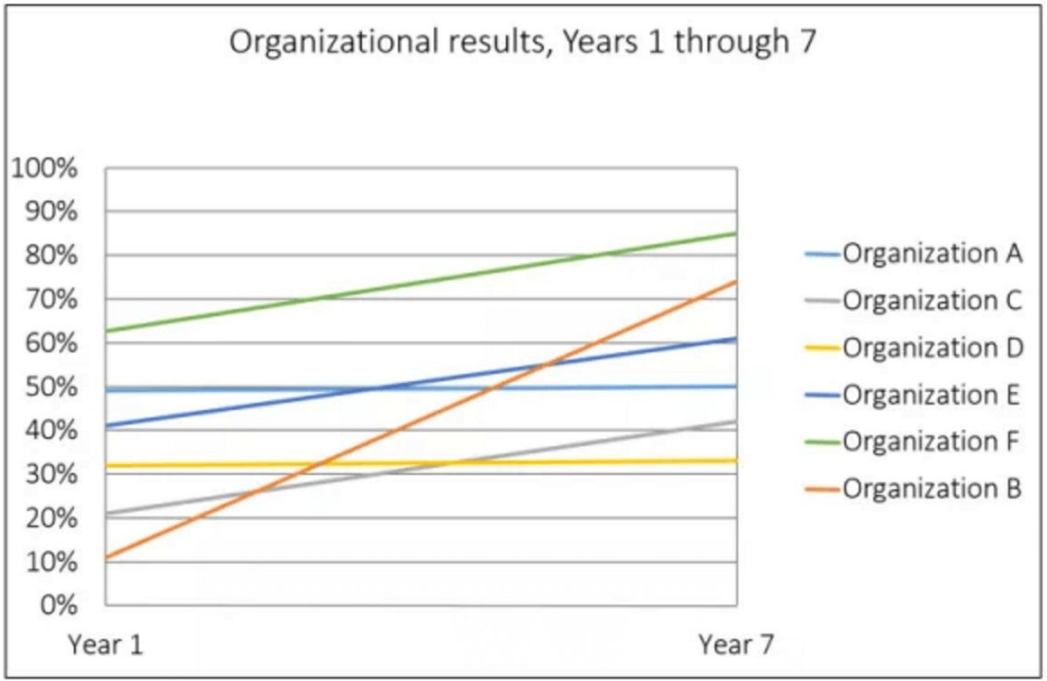
Case Study - Pie Charts

https://dataviztraining.dwpdata.info/img/graph.gif

Case Study - Map

https://dataviztraining.dwpdata.info/img/map.gif

Case Study - Annotation



Case Study - Annotation



First Release

16th January 2014

Coverage: Great Britain
Theme: Social and welfare

Fraud and Error in the Benefit System: 2012/13 Estimates (Great Britain)

Estimates of the levels of fraud and error in the benefit system in Great Britain are published today. These are available on the GOV.UK website at:

https://www.gov.uk/government/collections/fraud-and-error-in-the-benefit-system

The publication presents the 2012/13 estimates of total overpayments and underpayments across the benefit system in Great Britain. This release also covers individual estimates for Income Support, Jobseeker's Allowance, Pension Credit and Housing Benefit for 2012/13.

Headline results

Estimates of Fraud and Error across all benefits in 2012/13

The estimate of total overpayments due to fraud and error across all benefits is £3.5bn; this is 2.1% of the total benefit expenditure, which was £166.6bn in 2012/13.

However, the net loss to DWP is less than the total overpayments since DWP and Local Authorities identified and recovered around £900m of overpayments in 2012/13.

The estimate of total underpayments due to fraud and error across all benefits is £1.6bn; this is 0.9% of the total benefit expenditure in 2012/13.

Estimates of Fraud and Error by type of error in 2012/13

- 0.7%, or £1.2bn, of total benefit expenditure is overpaid due to fraud;
- 1.0%, or £1.6bn, of total benefit expenditure is overpaid due to claimant error;
- 0.5%, or £0.8bn, of total benefit expenditure is overpaid due to official error.

- 0.6%, or £1.0bn, of total benefit expenditure is underpaid due to claimant error;
- 0.3%, or £0.5bn, of total benefit expenditure is underpaid due to official error.

Estimates of Fraud and Error by individual benefits in 2012/13

Key findings for the individual benefits that are measured on a continuous basis are that:

- 4.3%, or £230m, of Income Support expenditure (£5.5bn) was overpaid;
- 4.1%, or £210m, of Jobseeker's Allowance expenditure (£5.2bn) was overpaid;
- 6.4%, or £490m, of Pension Credit expenditure (£7.6bn) was overpaid;
- 5.1%, or £1,220m, of Housing Benefit expenditure (£23.7bn) was overpaid.

- 1.3%, or £70m, of Income Support expenditure (£5.5bn) was underpaid;
- 0.4%, or £20m, of Jobseeker's Allowance expenditure (£5.2bn) was underpaid;
- 2.3%, or £180m, of Pension Credit expenditure (£7.6bn) was underpaid;
- 1.6%, or £380m, of Housing Benefit expenditure (£23.7bn) was underpaid.

Department for Work & Pensions

Issued by:
Information, Governance and Security Directorate
Fraud and Error Measurement Analysis
Department for work and Pensions

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Department for Work & Pensions

Fraud and Error in the Benefit System

Preliminary data for 2015/16

Biannual

Published: 19 May 2016
Great Britain

The Department for Work and Pensions administers welfare benefits to around 22 million people. 'Fraud and Error in the Benefit System' estimates **overpayments** (the total amount of money lost to the department because claimants are paid too much) and **underpayments** (the total amount of money lost to claimants who are not paid enough). This is the first year that we have measured fraud and error on Universal Credit. Further information on the change of categorisation of overpayments can be found in the Background and Methodology document.

Main stories

1.8% of expenditure overpaid

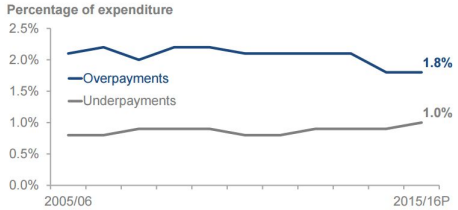
This remains the **lowest recorded rate**. It amounts to **£3.1bn** of overpayments.

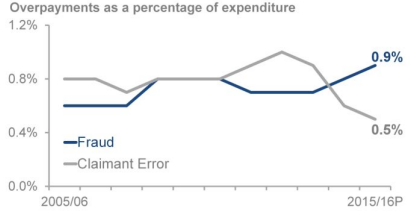
1.0% of expenditure underpaid

This is the **highest recorded rate**. It amounts to **£1.8bn** of underpayments.

We changed the way we categorise overpayments after 2013/14

This change in methodology means some overpayments have switched to being categorised as fraud rather than claimant error. This is driving recent trends: fraud is now at its **highest recorded rate**; claimant error is at its **lowest recorded rate**.

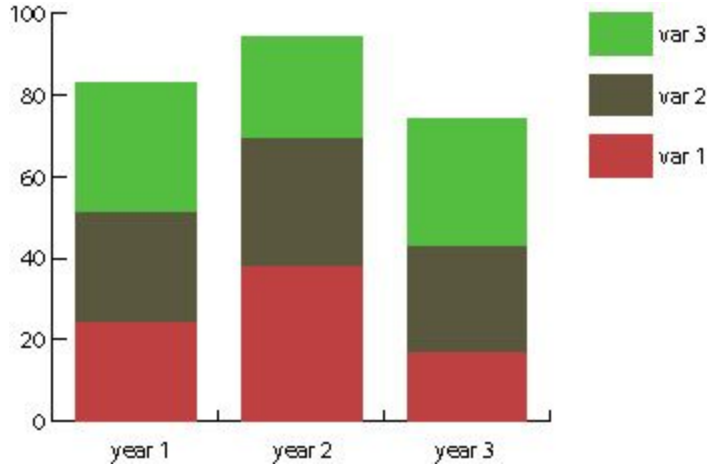




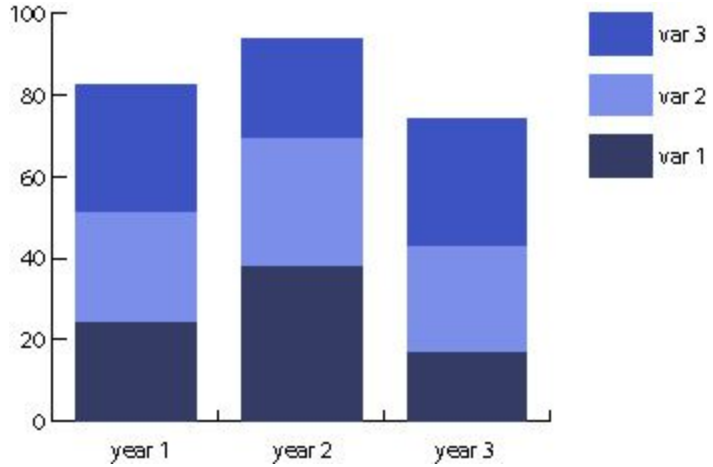
Example

https://chartaccent.github.io/

Case Study - Color



Case Study - Color



Think carefully before choosing a color scheme...

“Get it right in black and white”

Make sure your design works in greyscale before adding in color. Don’t rely on color alone to tell the story.

“The safest hue is blue”

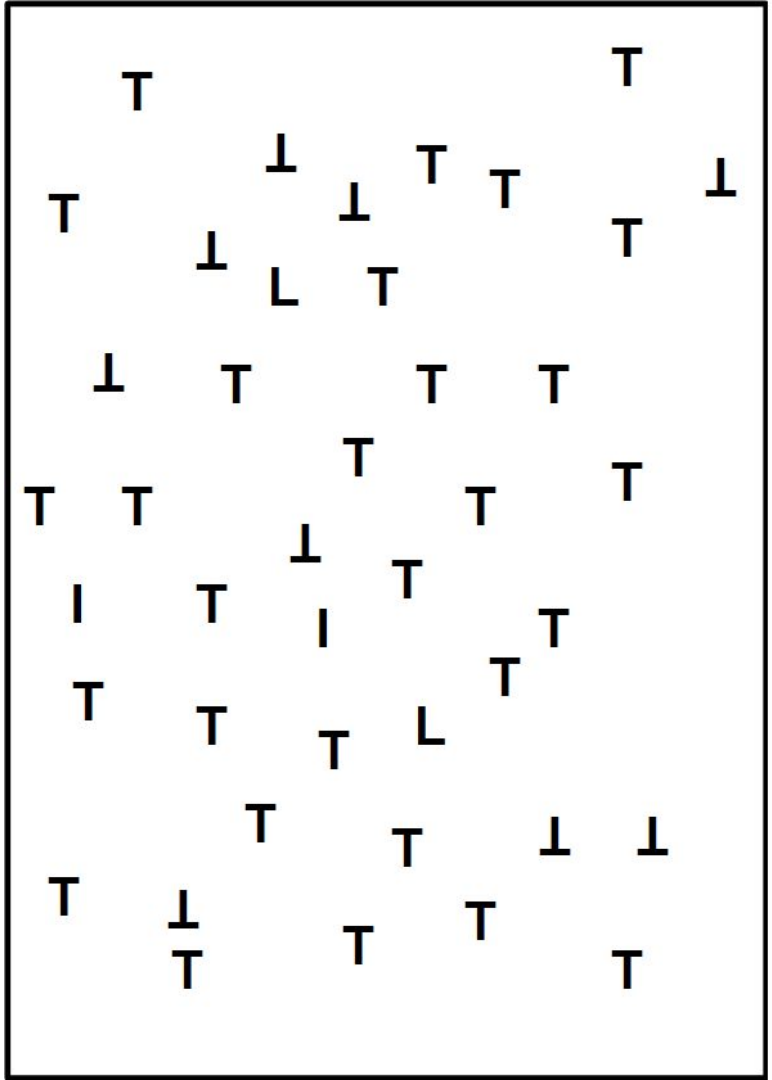
The richest color across all types of color blindness is blue, therefore blue is often a good color choice.

“Red and green should never be seen”

Difficult to distinguish from one another in the more common types of color blindness.

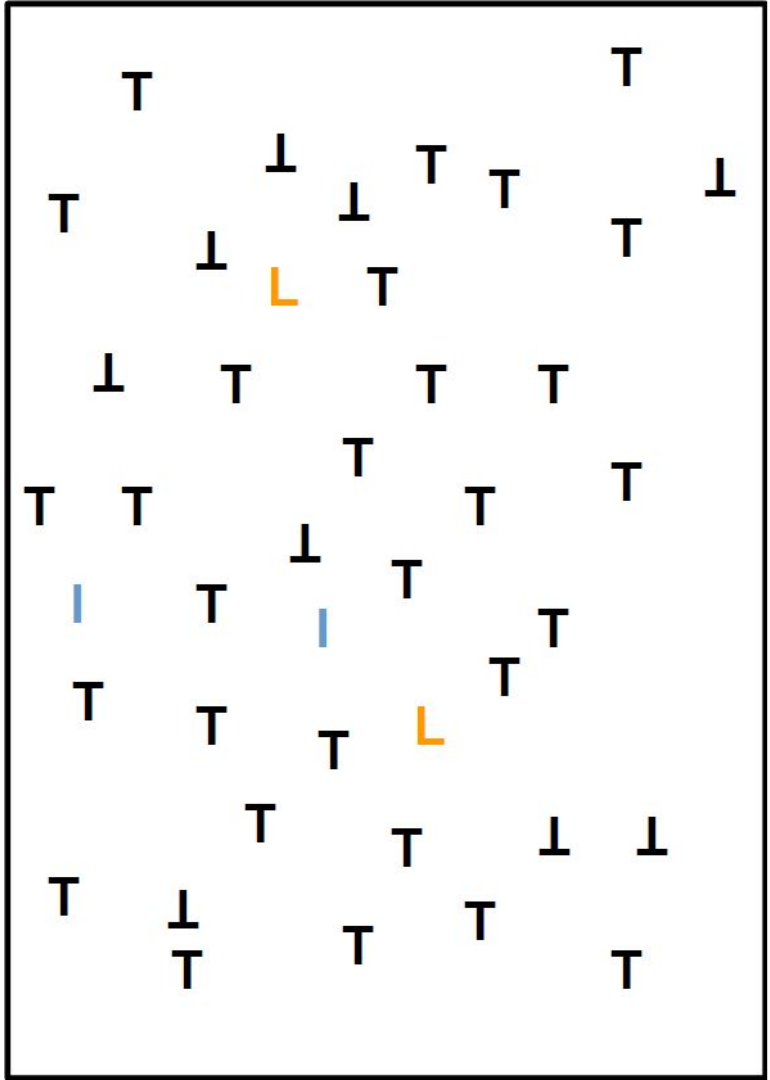
Case Study - Color

Can you find the Ls and Is among all the Ts?



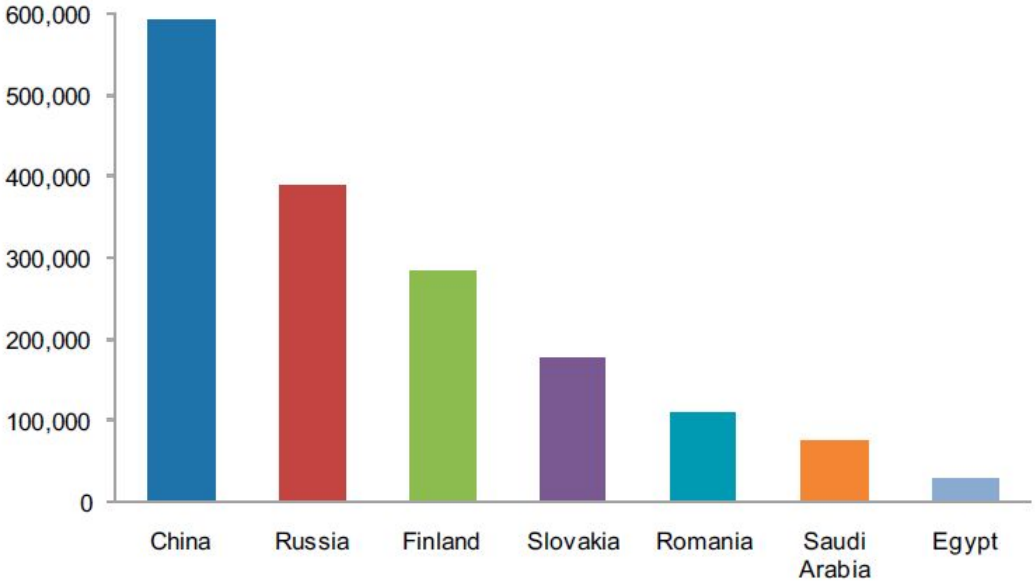
Case Study - Color

Can you find the Ls and Is among all the Ts?



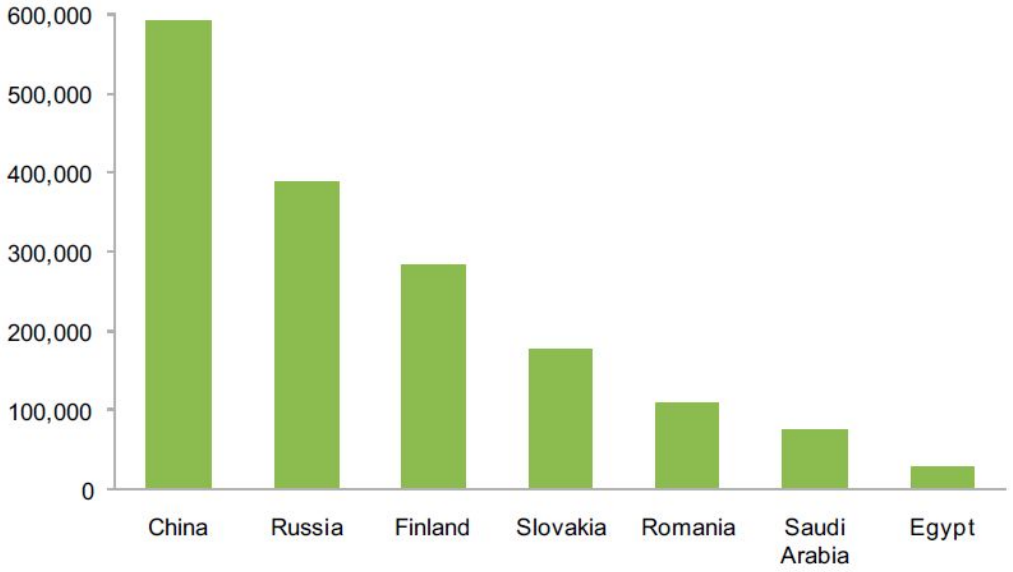
Case Study - Color

Only use different colours when they represent helpful differences in the data.



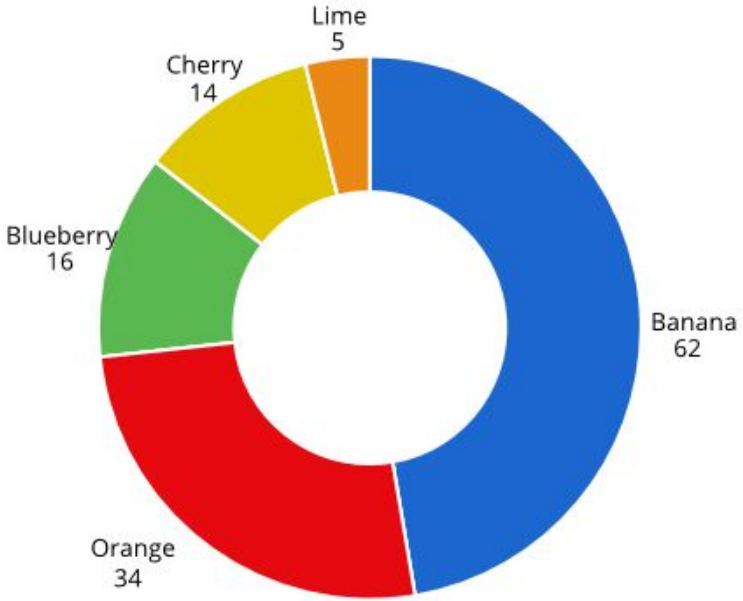
Case Study - Color

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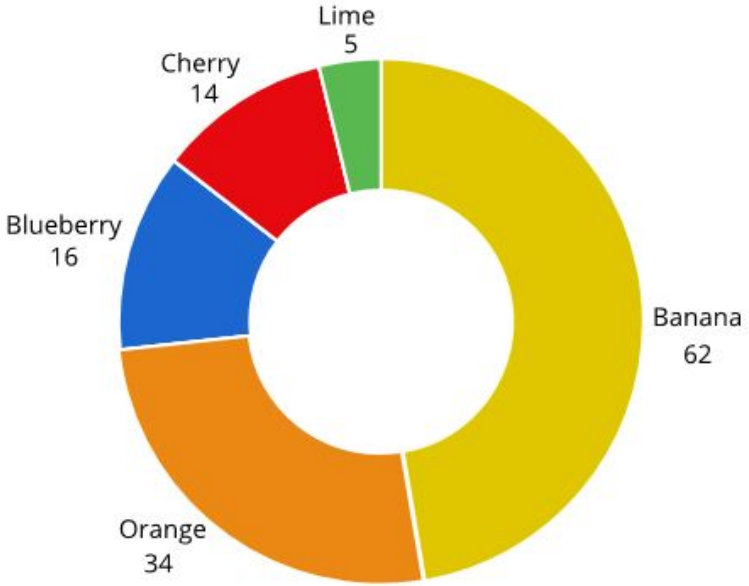
Case Study - Color

Some colours are associated with particular meaning.



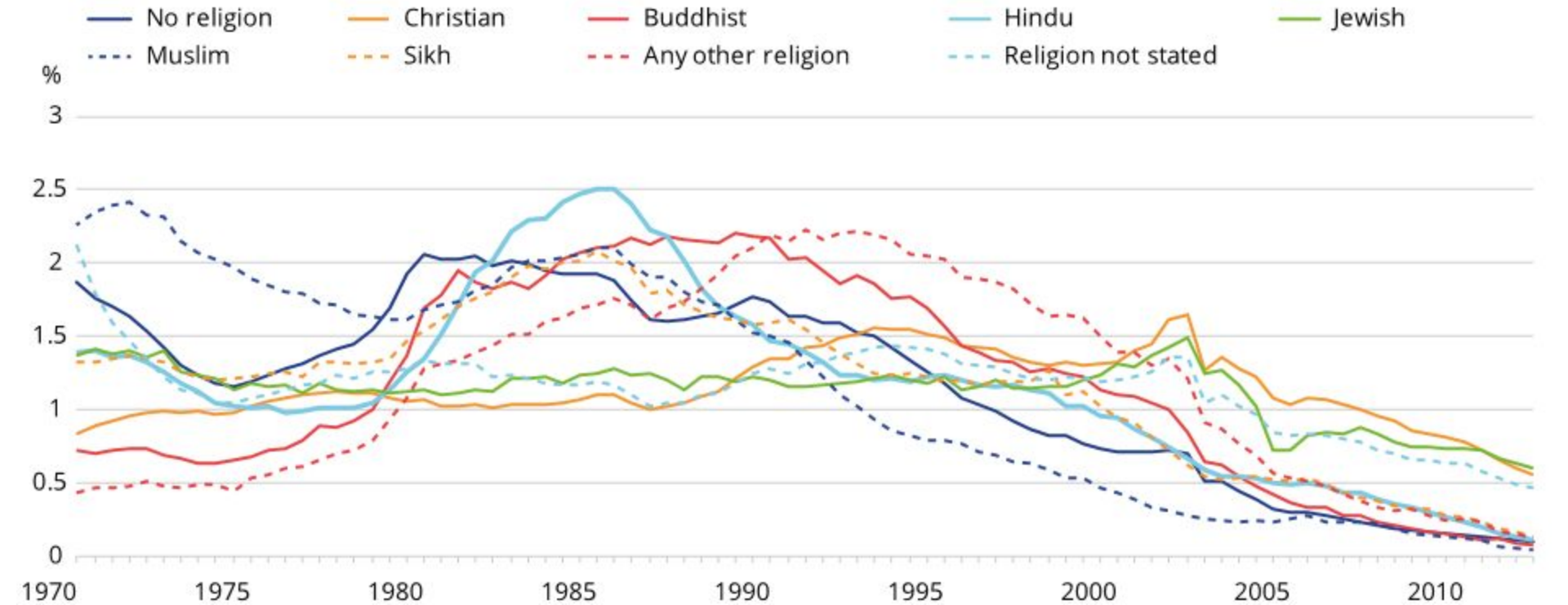
Case Study - Color

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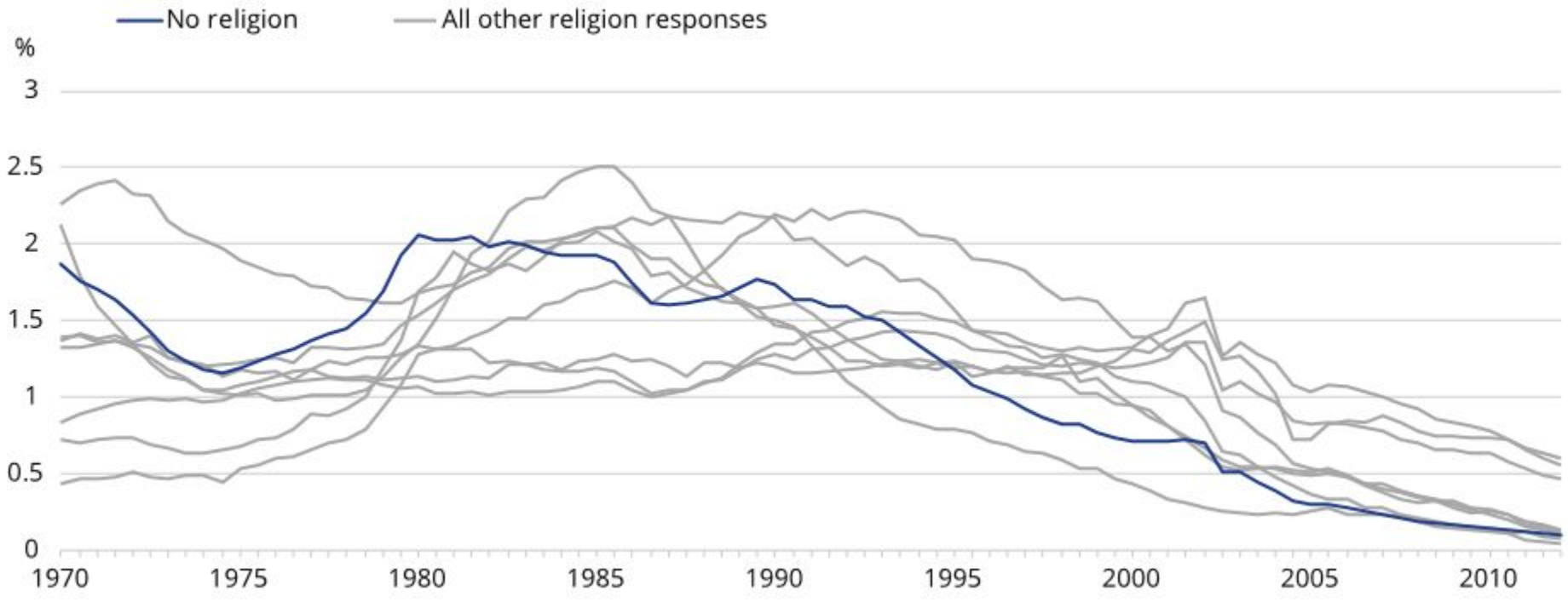
Case Study - Color

Crowded Charts



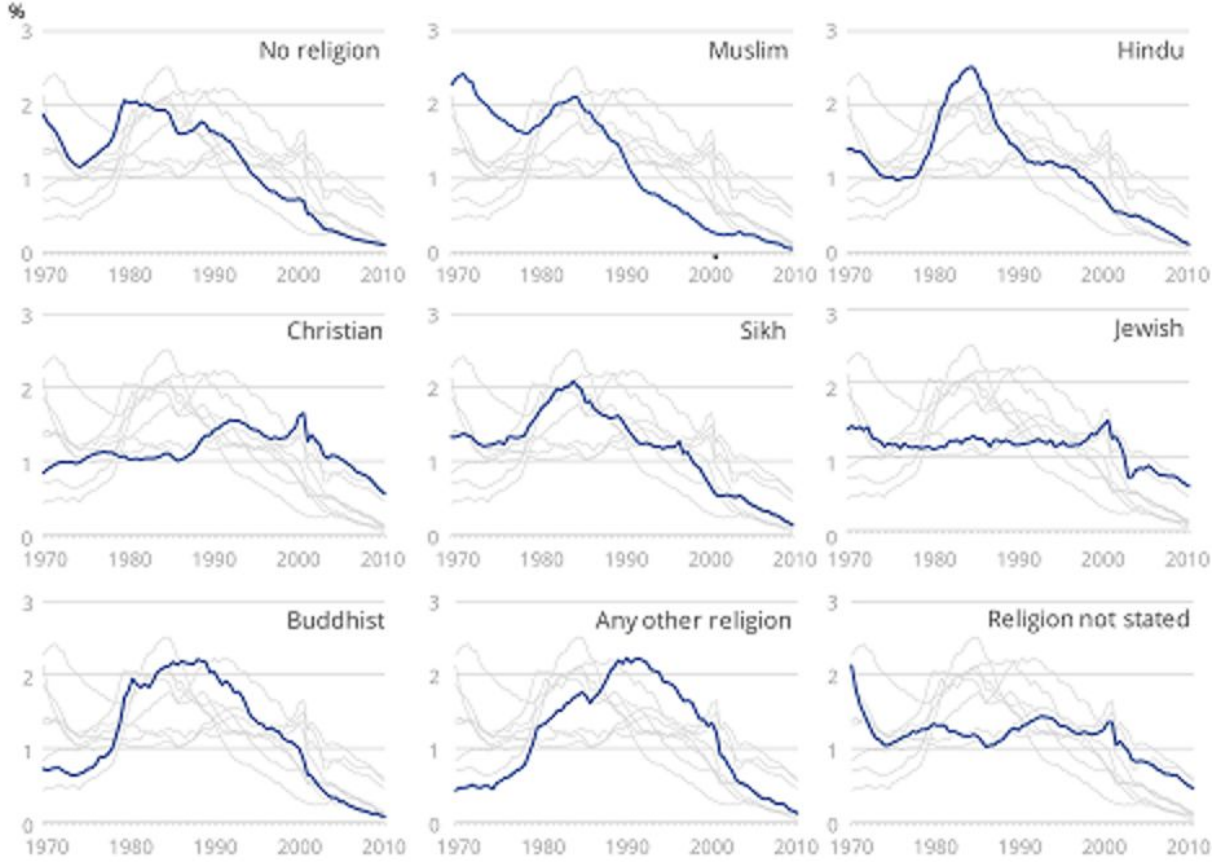
Case Study - Color

Crowded Charts



Case Study - Color

Crowded Charts



Q&A